Bounding a trained network's error by compressing it

Summary of the `compression-generalization` project, prompted by Kaarel Hänni. Written by Claude (Anthropic), 28 October 2026. Not reviewed by a human: proofs and numbers were checked by AI referees, and the follow-up's citations against the papers. Paper, follow-up note, code and records: `https://github.com/kaarelh/AI-works/tree/main/compression-generalization`.

The idea. If a network g with an L-bit description, in a code fixed before the data, agrees with the trained network f on most inputs, then with probability $1 - \delta_1 - \delta_2$

$$\mathrm{err}(f) \;\le\; \overline{\mathrm{Bin}}\big(k_S(g),\, m,\, \delta_1 2^{-L}\big) \;+\; \overline{\mathrm{Bin}}\big(j,\, N,\, \delta_2\big),$$

an Occam bound on g's $k_S(g)$ training errors plus a binomial bound on $\Pr[f \neq g]$ from j disagreements on N fresh inputs. No validation labels are needed. The questions: how to certify the second term, and how to find g.

Setting	Network f	Test error	Bound	Surrogate g
Known input distribution (Boolean, uniform on $\{\pm1\}^{50}$, $m = 16{,}000$)				
noise-free junta and threshold targets	MLP, 289k param.	0–0.001	0.002–0.005	28–98-bit table or student
10% label noise that f interpolates	same	0.14	0.17	the clean rule
dense targets (random ReLU teachers)	MLPs	0.02–0.07	0.11–0.21	best of 32 restarts
Any input distribution (20 bits, 3-parity, $m = 4000$)	MLP	0	0.0065	exact equivalence + patches
Unlabeled inputs (MNIST)	784-100-10	0.023	0.31	18k-bit coded network
twin trained on a fixed half of the data	784-1024-10	0.016	0.042	retraining, no search
Training set only, dithered transfer (MNIST)	784-8-10	0.070	0.44	weight-space certificate
pruned to a sparse DCT update, fine-tuned	784-100-10	0.046	0.31	same

Parameter-counting and VC bounds are 1.0 for the Boolean MLPs and the dense wide MNIST networks, but 0.71 (VC) for 784-8-10 and 0.62 (exact code) for the sparse network. The training-set-only bound is vacuous for wide dense networks (0.97 at 128 hidden units).

Main paper

- **Prior art.** The reduction with unlabeled inputs goes back to Kääriäinen (2005); Bazinet, Zantedeschi and Germain (UAI 2026) apply it to deep networks (1.8–4.8% on an MNIST CNN). Bounds on a compressed g alone (Arora et al. 2018; Lotfi et al. 2022) do not bound f.
- **Known D_X is free.** On Boolean tasks a 289k-parameter MLP compresses in function space to 28–98 bits; the bound nears test error as m grows and stays informative for partly learned networks (3-parity 0.30 vs 0.29; 6-junta 0.34 vs 0.17).
- **Any input distribution.** On 20-bit inputs, listing the inputs where f and its compressed network disagree gives bounds valid for every input distribution, if f is nearly exact: at most 8 exceptions below 10^{-5} test error (3-parity: 0.0065), but 72–178 and bounds 0.33–0.59 at $\approx 10^{-4}$.
- **Training set only.** New mechanism, a *dithered transfer theorem*; not tighter than Biggs and Guedj (2022) on erf/GELU nets. Compressing f by subtractive dithered quantization with public randomness makes $g - f$ independent noise, so a certificate computed from g alone transfers through the same Occam bound to f itself (proved for one-hidden-layer ReLU networks), without the $\sqrt{d}$ loss of worst-case perturbation sets. It needs the weights to the precision the margins demand, so it is non-vacuous only for narrow networks or ones pruned and fine-tuned to a sparse update for the bound (at higher test error).
- **Initialization costs nothing.** Codes may use any data-independent randomness; function-space codecs never refer to it. An init-kernel codec selects rank $1 + n$ for majority targets (lazy training, $n = 30$ only).
- **Open problem.** A training-set-only certificate that sees functional simplicity (f close to a 5-junta) instead of weight-space proximity.

Follow-up: can g be found efficiently?

- **Validity needs no search budget;** compute only affects tightness. With a short noise-free target, exhaustive search certifies every trained network to within $(L_t \ln 2 + \ln \frac{1}{\delta})/m$ of its error, plus sampling slack.
- **Worst case: hard.** If pseudorandom functions exist, no polynomial-time algorithm with queries to f finds an efficient g that beats chance on uniform inputs, although f is short (classical).
- **Boolean tasks: polynomial search sufficed.** Influence search recovers every junta and threshold target (31–98 bits) in seconds to minutes. On dense targets (signs of random 4- and 8-unit ReLU networks), single runs reach 1.24–1.92 times the bound the true teacher gives, because small students get stuck; 32 restarts of a teacher-width student (a width sweep recovered the width on the 2 networks tried), trained longer and quantized finer, reach 0.96–1.11 times, in about 12 CPU-minutes per network.
- **Subsampling hidden units** (Maurey: $\mathbb{E}\|g - f\|^2 = (A^2 - \|f - c\|^2)/k$, A the path norm) works when few units carry f: 2 units on mean-field networks for majority targets, but 16–32 on standard ones (bounds 0.07–0.30). Alone it was never the best search.
- **Subsampling the data works better:** a twin trained on a fixed half, tested on the other half and compared with f on unlabeled inputs, gave 3–7 times tighter bounds than compression on all four MNIST/Fashion MLPs (Bazinet et al.'s 1.8% uses the same construction). It loses near the learning threshold, on the 50-bit majority and with interpolated label noise.
- **Is f close to a short program?** Yes by definition for short targets; the dense teachers are long relative to m; on MNIST we found no short g close to f (best: 6–12k bits, 0.255). Tiny 697- to 1,936-parameter MNIST networks reported at 99% suggest one may exist; untested.

Caveats. Mostly one seed per network; small Boolean tasks; MLPs only on images; codecs tuned on the same data (charged 4 bits and $\delta/16$); all checking by AI, not humans. Follow-up citations were checked against the papers (Springer, SIAM, AAAI ones via later papers); main-paper related work rests partly on search snippets.